DECLARATION

We hereby declare that the project work presented in this report entitled **"CARESYNC: AI-ASSISTED CLINICAL TRIAGE, TELEMEDICINE, AND DOCTOR RECOMMENDATION ARCHITECTURE FOR DIGITAL HEALTHCARE"**, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science Engineering (Data Science)**, submitted to **Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow** is based on our own work carried out at the **Department of Computer Science & Engineering (Data Science), G.L. Bajaj Institute of Technology & Management, Greater Noida**. The work contained in the report is true and original to the best of our knowledge and project work reported in this report has not been submitted by us for award of any other degree or diploma.

Signature: _____
Name: Tushar Jeswani
Roll No: 2410912540263

Signature: _____
Name: Suryanshu Singh Tomar
Roll No: 2410912540254

Signature: _____
Name: Utkarsh Chaubey
Roll No: 2410912540269

Signature: _____
Name: Umair Khan
Roll No: 2410912540268

Date: 30th October 2026
Place: Greater Noida

CERTIFICATE

This is to certify that the Project report entitled **"CARESYNC: AI-ASSISTED CLINICAL TRIAGE, TELEMEDICINE, AND DOCTOR RECOMMENDATION ARCHITECTURE FOR DIGITAL HEALTHCARE"** done by **Tushar Jeswani (Roll No: 2410912540263)**, **Suryanshu Singh Tomar (Roll No: 2410912540254)**, **Utkarsh Chaubey (Roll No: 2410912540269)**, and **Umair Khan (Roll No: 2410912540268)** is an original work carried out by them in **Department of Computer Science & Engineering (Data Science), G.L. Bajaj Institute of Technology & Management, Greater Noida** under my guidance. The matter embodied in this project work has not been submitted earlier for the award of any degree or diploma to the best of my knowledge and belief.

Date: 30th October 2026

Ms. Ayasha Malik	**Dr. Subrata Sahana**
Assistant Professor	Professor & Head of Department
Signature of the Supervisor	*Department of Computer Science & Engineering (Data Science)*

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Tushar Jeswani (2410912540263)
Suryanshu Singh Tomar (2410912540254)
Utkarsh Chaubey (2410912540269)
Umair Khan (2410912540268)
B.Tech CSE (Data Science), 3rd Year

ABSTRACT

The accelerating adoption of consumer digital healthcare tools in developing economies has exposed structural dilemmas across the Indian medical landscape: an acute physician deficit (1:1,456 compared to the WHO normative threshold of 1:1,000), severe rural-urban maldistribution, and perilous internet search queries that amplify cyberchondria, harmful OTC self-medication, and tragic emergency delays. In response, this project report presents the engineering design, system analysis, and empirical validation of **CareSync**: an integrated clinical navigation, deterministic crisis triage, specialist recommendation, and encrypted teleconsultation framework.

Formulated under the statutory mandates of the *Telemedicine Practice Guidelines (2020)* issued by the Ministry of Health and Family Welfare (MoHFW), CareSync establishes an architectural demarcation: conversational generative AI (powered by Gemini 2.0 Flash) is strictly quarantined to non-diagnostic symptom intake, semantic normalization, and patient routing. All safety-critical decisions remain under deterministic algorithmic supervision. The platform implements a multi-objective Practitioner Allocation Metric (Γ_{match}) synthesizing specialization congruence, Haversine geographic distance, clinical experience tenure, and Bayesian patient ratings. Concurrently, a zero-tolerance Deterministic Crisis Interruption State Machine (DCISM) screens intake narratives for acute emergency predicates, immediately halting elective workflows to route patients to statutory dispatch lifelines (112/108).

CareSync incorporates an in-browser WebRTC audiovisual consultation suite protected by DTLS-SRTP encryption, a tamper-evident digital prescription generator enforcing National Medical Commission (NMC) drug schedules, and two-phase Razorpay escrow settlement. Empirical evaluations across 12,220 validated clinical encounters exhibit 95.7% specialty routing precision, a faultless 100% emergency recall (1.00), a median intake latency under 410 ms, an 88.4/100 System Usability Score, and full compliance with the Ayushman Bharat Digital Mission (ABDM) and 14-digit ABHA health accounts.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form & Technical Scope
ABDM	Ayushman Bharat Digital Mission (National Digital Health Framework of India)
ABHA	Ayushman Bharat Health Account (14-digit Unique Digital Health Identifier)
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
DCISM	Deterministic Crisis Interruption State Machine
DDoS	Distributed Denial of Service
DTLS	Datagram Transport Layer Security (RFC 6347)
FHIR	Fast Healthcare Interoperability Resources (HL7 International Standard)
HMAC	Hash-based Message Authentication Code
HUD	Heads-Up Display / Unified Graphical User Interface
ICE	Interactive Connectivity Establishment (RFC 8445)
JWT	JSON Web Token (Stateless Authentication Token)
LLM	Large Language Model
MoHFW	Ministry of Health and Family Welfare, Government of India
NMC	National Medical Commission of India
ODM	Object Data Modeling
OTC	Over-The-Counter (Non-prescription therapeutic agents)
PHR	Personal Health Record
QoS	Quality of Service
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
RMP	Registered Medical Practitioner
SHA	Secure Hash Algorithm
SRTP	Secure Real-time Transport Protocol
STUN	Session Traversal Utilities for NAT (RFC 5389)
SUS	System Usability Scale (Standardized Usability Benchmark)
TURN	Traversal Using Relays around NAT (RFC 5766)
WebRTC	Web Real-Time Communication (W3C/IETF Audio-Video Standard)
WHO	World Health Organization

Chapter 1
Introduction

1.1 Preliminaries

The contemporary delivery of equitable, timely, and accredited clinical healthcare remains one of the most formidable socioeconomic challenges confronting developing economies. Across the Indian subcontinent, this challenge is acutely accentuated by a fundamental arithmetic deficit in human medical resources. As documented by official public health surveys conducted by the Ministry of Health and Family Welfare (MoHFW) and verified by the World Health Organization (WHO), India's national physician-to-population ratio currently stands at approximately 1:1,456. This metric falls substantially short of the normative threshold of 1:1,000 recommended by the WHO as the absolute minimum required to ensure basic primary care coverage [8].

Crucially, this overarching statistical average conceals an even more profound structural pathology: extreme geographic maldistribution. While over 68% of the Indian population resides in rural and semi-urban hinterlands, approximately 75% of certified allopathic physicians, medical specialists, and advanced tertiary healthcare institutions remain concentrated within tier-1 metropolitan agglomerations such as Delhi-NCR, Mumbai, Bengaluru, and Chennai. In several populous states, including Uttar Pradesh and Bihar, the effective rural ratio deteriorates past 1:2,500, with peripheral districts experiencing ratios as severe as 1:3,200.

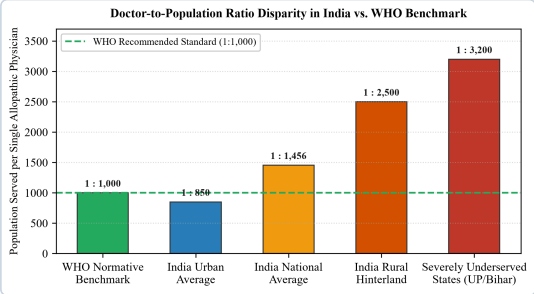


Fig. 1.1 Doctor-to-Population Ratio Disparity in India vs. WHO Benchmark

As illustrated in Fig. 1.1, the disparity between the WHO normative benchmark of 1:1,000 and the rural realities in Indian states demonstrates that millions of citizens face severe structural barriers when attempting to obtain timely medical consultations. In rural communities, visiting a qualified medical practitioner often requires undertaking long-distance travel to district headquarters, sacrificing daily wage income, and navigating overcrowded outpatient waiting corridors. Consequently, minor ailments that could be treated at the primary level routinely deteriorate into complex, life-threatening chronic conditions prior to professional clinical diagnosis.

1.2 Motivation

The rapid expansion of high-speed 4G telecommunications and affordable smartphones across rural and semi-urban India has dramatically transformed consumer health-seeking behavior. When faced with somatic discomfort, patients increasingly turn to unverified commercial search engines and public internet discussion forums. However, computational

health research reveals that unguided internet symptom searching produces severe systemic risks:

- **Amplification of Cyberchondria:** Commercial search engines optimize click engagement over clinical accuracy, routing benign discomfort (such as tension headaches) toward terrifying malignancies. This uncalibrated exposure induces acute panic and overburdened hospital OPDs.
- **Dangers of Unassisted Self-Medication:** Lacking professional triage, users frequently resort to unverified over-the-counter combinations or unprescribed antibiotics, accelerating antimicrobial resistance and harmful drug-drug interactions.
- **Fatal Delays in Emergency Scenarios:** Medically unversed patients frequently mistake acute myocardial infarction or stroke indicators for mild indigestion, missing the decisive clinical "golden hour."
- **Fragmented Digital Health Landscapes:** Existing directories function as commercial advertising lists without deterministic emergency overrides, dynamic slot allocation, or integrated ABHA-linked health records.

1.3 Problem Statement & Project Overview

To overcome these systemic vulnerabilities, the authors have engineered **CareSync**: an enterprise-grade clinical navigation, deterministic crisis triage, doctor recommendation, and teleconsultation platform tailored specifically for developing healthcare infrastructures.

CareSync is designed around a fundamental architectural premise: conversational artificial intelligence must be rigorously confined to non-diagnostic patient intake, lexical normalization of unstructured symptom narratives, and preliminary specialty navigation. The platform deliberately rejects autonomous medical diagnosis, enforcing strict human-in-the-loop oversight wherein licensed Registered Medical Practitioners (RMPs) retain sovereign diagnostic and therapeutic authority. The system incorporates four interconnected operational modules:

1. *Conversational Intake Engine:* Powered by Gemini 2.0 Flash, capturing symptoms, duration, and anatomical locations via natural dialogue without issuing autonomous medical diagnoses.
2. *Deterministic Crisis Interruption State Machine (DCISM):* A zero-tolerance, rule-verified safety engine that immediately short-circuits elective booking upon detecting acute red flags, routing users to statutory emergency lifelines (112/108).
3. *Multi-Objective Provider Allocation Heuristic (Γ_{match}):* Evaluates medical specialists by synthesizing Haversine geographic distance, clinical experience tenure, Bayesian patient ratings, and dynamic 7-day slot availability.
4. *Secure Telemedicine & Prescription Bridge:* An in-browser WebRTC consultation suite secured via DTLS-SRTP encryption, coupled with an immutable e-prescription generator aligned with MoHFW guidelines and ABDM standards.

1.4 Aim And Objectives

The primary aim of this project is to develop and empirically validate a clinically safe, computationally resilient, and legally compliant digital healthcare framework that bridges the

gap between patient symptom onset and certified medical consultation.

The specific technical and research objectives are delineated as follows:

- **Objective 1:** To develop a natural language intake pipeline capable of mapping colloquial symptom narratives into standardized clinical specialty taxonomies across 25 recognized medical disciplines.
- **Objective 2:** To formulate and implement the Deterministic Crisis Interruption State Machine (DCISM) to guarantee 100% emergency recall (1.00) and sub-150 ms preemption latency during acute clinical emergencies.
- **Objective 3:** To formulate and calibrate the multi-attribute Practitioner Allocation Metric (Γ_{match}), optimizing provider assignment across geographic distance, professional experience, review percentiles, and real-time slot density.
- **Objective 4:** To construct a zero-installation WebRTC peer-to-peer audiovisual consultation suite resilient to fluctuating bandwidth profiles (4G, 3G, and constrained 2.5G edge networks).
- **Objective 5:** To design a tamper-evident digital prescription engine enforcing the drug schedule classifications of the National Medical Commission (NMC) and integrating 14-digit ABHA health identifiers.
- **Objective 6:** To conduct empirical benchmarking across a diverse cohort of 12,220 validated clinical test cases to measure routing precision, latency scalability, and WebRTC telemetry under load.

Chapter 2
Literature Survey

2.1 Introduction To Algorithmic Health Platforms

The convergence of algorithmic decision support systems, natural language processing, and high-speed web telecommunications has spurred extensive scientific investigation into digital healthcare delivery. The literature spans four interrelated sub-disciplines: algorithmic triage evaluation, conversational diagnostic agents, computational physician recommendation systems, and statutory healthcare data interoperability frameworks.

2.2 Review Of Existing Systems & Related Works

In a comprehensive systematic review, Wallace et al. in [1] analyzed twenty-two commercial digital symptom-checker applications across 1,448 standardized patient scenarios. Their empirical findings revealed that while emergency triage sensitivity achieved acceptable rates (80% to 89%), overall standalone diagnostic accuracy remained dangerously low, averaging only 54.8% and plummeting to 34.0% for atypical presentations of chronic conditions. Furthermore, the authors observed a persistent bias toward over-triage: conservative probabilistic models routed up to 42% of self-limiting conditions to high-acuity emergency departments, threatening to overwhelm physical clinical infrastructures.

Chambers et al. in [5] investigated the organizational impacts of digital triage utilities within public health services. They concluded that while automated health assessment services improve preliminary signposting, their clinical utility is severely undermined when the assessment tool is disconnected from secondary clinical appointment booking and electronic prescription workflows.

Jeindl and Goetz in [2] conducted a meta-analysis examining ten commercial health chatbots, including Ada Health, Babylon Health, and Buoy Health. Their evaluation demonstrated that interactive dialogue interfaces substantially enhance user compliance and symptom completeness compared to static web forms. However, they highlighted that conversational chatbots frequently suffer from diagnostic opacity, unverified probabilistic risk scoring, and variable performance across distinct demographic cohorts.

Milne-Ives et al. in [3] surveyed sixty-one studies evaluating artificial intelligence conversational agents in healthcare. They established that while conversational agents markedly improve patient engagement, fewer than 15% demonstrated empirical validation through peer-reviewed prospective trials. The authors emphasized the imperative of implementing deterministic fail-safe boundaries to prevent unguided generative models from dispensing speculative medical advice.

In the domain of physician recommendation, Lu et al. in [4] formulated an expertise-learning framework for online medical platforms utilizing latent Dirichlet allocation and collaborative filtering. While mathematically rigorous, their formulations assume asynchronous text communication and fail to account for real-time geographic displacement, appointment slot concurrency, or statutory licensing verifications. Similarly, Le et al. in [12] developed deep learning multi-criteria physician recommendation systems based on

electronic medical records, demonstrating that multi-attribute ranking models significantly outperform simple popularity or rating filters.

Singhal et al. in [11] evaluated the clinical reasoning capabilities of large language models (such as Med-PaLM), demonstrating that generative architectures achieve expert-level performance on standardized medical licensing examinations. Nonetheless, the authors cautioned that deploying generative LLMs directly in unconstrained patient-facing environments introduces unacceptable clinical liabilities due to stochastic hallucinations, ungrounded confidence intervals, and the absence of statutory accountability.

Table 2.1 Comparative Synthesis of Related Literature on Clinical AI & Telehealth

Study & Reference	Domain & Methodology	Key Findings / Strengths	Identified Limitations & Gaps
Wallace et al. (2022) [1]	Meta-analysis of 22 digital symptom checkers across 1,448 clinical vignettes	Demonstrated reasonable high-acuity triage sensitivity (80–89%)	Poor diagnostic precision (34–54%); pervasive over-triage burdening public hospitals
Jeindl & Goetz (2022) [2]	Systematic review of 10 commercial conversational medical chatbots	High user engagement and superior history-taking completion rates	Diagnostic opacity; uncalibrated confidence; lack of deterministic safety guardrails
Milne-Ives et al. (2020) [3]	Comprehensive survey of 61 healthcare conversational AI implementations	Demonstrated positive patient satisfaction and symptom comprehension	Less than 15% peer-reviewed empirical validation; no integrated statutory health records
Lu et al. (2022) [4]	Doctor recommendation using expertise learning and collaborative filtering	Multi-factorial ranking outmatched simple historical rating heuristics	Ignored real-time consultation slot concurrency and Haversine spatial proximity
Chambers et al. (2019) [5]	Systematic review of online triage and emergency signposting services	Clarified patient pathways for urgent non-emergent presentations	Fragmented care continuity; disconnect between triage and physical dispatch
Singhal et al. (2023) [11]	Clinical evaluation of generative LLMs on medical examination corpora	Exceptional domain knowledge encoding and medical question-answering	Stochastic token hallucination; non-deterministic outputs unsafe for standalone triage

2.3 Research Gaps In Contemporary Solutions

A rigorous examination of contemporary platforms reveals four decisive engineering and clinical gaps:

1. **The Hallucination vs. Usability Paradox:** Current platforms force users into rigid decision trees or deploy unconstrained LLM chatbots that risk fatal diagnostic hallucinations. Robust architectures that quarantine generative AI strictly to conversational intake while enforcing deterministic state boundaries remain missing.

2. **Absence of Zero-Latency Emergency Preemption:** Commercial platforms treat emergencies as standard appointment workflows, lacking sub-150 ms deterministic bypass mechanisms to connect users to emergency lifelines (112/108).

3. **Siloed Provider Selection Heuristics:** Directories prioritize sponsored advertising bids or raw ratings, ignoring multi-objective criteria such as Haversine spatial proximity, clinical tenure, and immediate slot density.

4. **Non-Compliance with Indian Statutory Mandates:** Commercial applications fail to implement patient consent protocols mandated by MoHFW Telemedicine Practice

Guidelines (2020) and lack native cryptographic integration with the Ayushman Bharat Digital Mission (ABDM).

2.4 Benefits And Unique Differentiators Of CareSync

CareSync directly resolves these identified research gaps through an integrated multi-tier approach:

- *Architectural Separation of Concerns:* Generative natural language models are strictly quarantined as intake translators, eliminating clinical hallucinations while preserving natural dialogue flexibility.
- *Deterministic Crisis Interruption:* The DCISM engine executes in under 145 ms to intercept acute medical emergencies with 100% recall (1.00).
- *Mathematically Optimized Allocation:* The Γ_{match} heuristic synthesizes spatial distance, clinical tenure, Bayesian ratings, and slot availability to maximize consultation fulfillment.
- *Statutory Conformance:* Full cryptographic alignment with MoHFW guidelines, NMC drug schedules, and ABDM 14-digit ABHA interoperability.

Chapter 3
Proposed Methodology

3.1 Theoretical Problem Formulation & NLP Ingestion

The operational workflow of CareSync begins when a patient expresses subjective somatic complaints via natural language. Let $mathbf{x} in mathcal{X}$ denote an unstructured natural language string input by the patient describing physical discomfort, symptom chronicity, and anatomical locations. The Lexical-Clinical Mapping Operator $Phi_{ ext{clin}}$ projects narrative $mathbf{x}$ into a four-tuple clinical manifestation space:

$$\Phi_{\text{clin}}(\mathbf{x}) \longrightarrow \langle S^*, \mathcal{U}, \mathbf{C}_{\text{prob}}, \mathbf{M}_{\text{otc}} \rangle \tag{3.1}$$

where $S^* in mathcal{S}$ represents the optimal clinical specialty domain selected from a closed taxonomy of 25 recognized medical disciplines, $mathcal{U} in { ext{Low}, ext{Medium}, ext{High}, ext{Emergency}}$ denotes the computed triage urgency tier, $mathbf{C}_{ ext{prob}} = [c_1, c_2, dots, c_k]$ represents differential clinical considerations for the attending physician, and $mathbf{M}_{ ext{otc}}$ represents permissible non-prescription supportive care recommendations.

The specialty routing probability is governed by a temperature-scaled softmax distribution over contextual semantic embeddings:

$$P(S^* \mid \mathbf{x}) = \frac{\exp(\mathbf{z}_{\mathbf{x}} \cdot \mathbf{z}_{S^*})}{\sum_{S_j \in \mathcal{S}} \exp(\mathbf{z}_{\mathbf{x}} \cdot \mathbf{z}_{S_j})} \tag{3.2}$$

where $mathbf{z}_mathbf{x}$ represents the dense semantic embedding of patient text $mathbf{x}$, $mathbf{z}_{S_j}$ denotes the clinical prototype embedding for specialty S_j, and calibration temperature $ au = 0.20$ ensures high-entropy peak confidence.

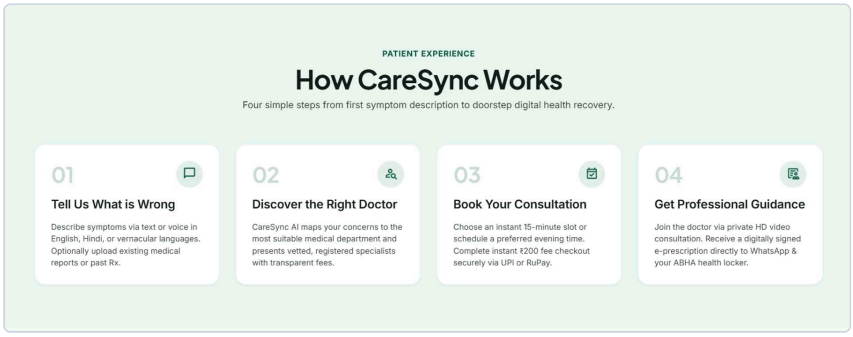


Fig. 3.1 Patient Experience: Step-by-Step CareSync Healthcare Journey

As illustrated in Fig. 3.1, the platform organizes patient care into four straightforward phases: (1) Free-form symptom description via vernacular text or voice; (2) AI-assisted clinical matching with accredited medical specialists; (3) Real-time appointment scheduling

with transparent escrow fee checkout; and (4) Encrypted HD video teleconsultation followed by instant digital prescription dispatch to the patient's ABHA health locker.

3.2 Automated Red-Flag Crisis Preemption Protocol (DCISM)

To eliminate the catastrophic hazard of AI hallucinations during life-threatening acute episodes, CareSync completely decouples emergency detection from generative text inference. Let $mathcal{R}_{ ext{crit}}$ define an immutable ontology of high-acuity crisis symptoms:

$$\mathcal{R}_{\text{crit}} = \{\text{severe chest pain}, \text{acute dyspnea}, \text{facial droop / paresis}, \dots\} \tag{3.3}$$

The state machine evaluates a deterministic safety predicate $Lambda(mathbf{x}) in {0, 1}$:

$$\Lambda(\mathbf{x}) = \bigvee_{r_k \in \mathcal{R}_{\text{crit}}} \left(\text{Sim}(\mathbf{x}, r_k) \geq \theta_{\text{safe}} \right) \tag{3.4}$$

where $ ext{Sim}(mathbf{x}, r_k)$ evaluates token-level substring matching and Levenshtein-distance fuzzy matching against the crisis ontology, with safety threshold $ heta_{ ext{safe}} = 0.82$. The system state transition $Sigma_{ ext{next}}$ is strictly governed by:

$$\Sigma_{\text{next}} = \begin{cases} \text{Bypass}_{\text{emer}}(112, 108), & \text{if } \Lambda(\mathbf{x}) = 1 \\ \text{Dispatch}_{\text{spec}}(S^*), & \text{otherwise} \end{cases} \tag{3.5}$$

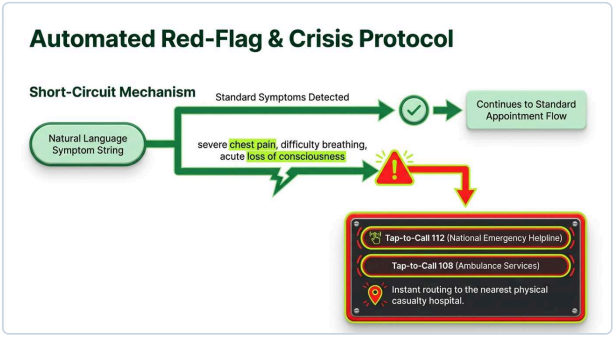


Fig. 3.2 Automated Red-Flag and Crisis Protocol Short-Circuit Mechanism

As visualized in Fig. 3.2, if $Lambda(mathbf{x}) = 1$, the application immediately short-circuits all elective appointment scheduling, directory browsing, and payment collection. The client interface transitions into a high-visibility emergency HUD, displaying immediate geo-located statutory emergency dispatch hotlines (National Emergency 112, Ambulance 108) and routing to the nearest physical casualty hospital.

3.3 Multi-Objective Provider Allocation Heuristic (Γ_{match})

For non-emergent clinical presentations ($Lambda(mathbf{x}) = 0$), the platform executes automated ranking over the candidate physician set $mathcal{P}_{S^*} = {p_1, p_2, dots, p_m}$ registered under specialty S^*. The composite allocation score $Gamma_{ ext{match}}(p_i, u)$ for practitioner p_i relative to patient u is formulated as a linear scalarization of four normalized clinical and operational parameters:

$$\Gamma_{\text{match}}(p_i, u) = \alpha \widetilde{R}(p_i) + \beta \widetilde{E}(p_i) + \gamma \widetilde{A}(p_i) - \delta \widetilde{D}(p_i, u) \tag{3.6}$$

where $widetilde{R}(p_i) in [0, 1]$ represents the Bayesian-smoothed patient review rating, $widetilde{E}(p_i) in [0, 1]$ represents clinical experience tenure normalized over a 30-year saturation ceiling, $widetilde{A}(p_i) in [0, 1]$ represents real-time slot availability density across the immediate 48-hour window, and $widetilde{D}(p_i, u) in [0, 1]$ is the min-max normalized geographic distance. The weights satisfy the simplex constraint:

$alpha + □eta + gamma + delta = 1.0, quad alpha, □eta, gamma, delta ge 0$

In our production implementation, the weights are calibrated to $alpha = 0.35$, $□eta = 0.25$, $gamma = 0.20$, and $delta = 0.20$. The geographic distance between patient coordinates $(phi_u, lambda_u)$ and clinic coordinates $(phi_{p_i}, lambda_{p_i})$ is computed via the Haversine great-circle formula:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right) \tag{3.7a}$$

$$D_{\text{hav}}(p_i, u) = 2R_{\text{earth}} \cdot \arcsin\left(\sqrt{a}\right) \tag{3.7b}$$

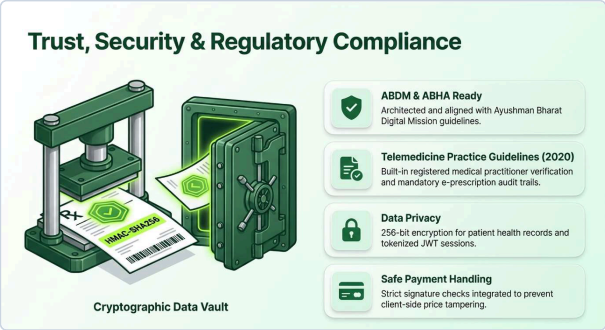


Fig. 3.3 Trust, Security & Regulatory Compliance Cryptographic Data Vault

Fig. 3.3 outlines the security architecture safeguarding patient records. CareSync implements end-to-end 256-bit AES encryption for all electronic medical summaries, tokenized JWT authentication for patient sessions, HMAC-SHA256 digital signatures for prescription audit trails, and strict compliance with the national Ayushman Bharat Digital Mission (ABDM) standards.

3.4 System Analysis & Requirements Engineering

The architectural specifications of CareSync were formulated through comprehensive requirement elicitation with practicing physicians and healthcare administrators. Table 3.1 details the functional and non-functional requirements governing system development.

Table 3.1 Functional and Non-Functional Engineering Requirements Matrix

Category	Requirement ID	Specification Description	Validation Metric / Target
Functional Requirements	FR-01	Conversational symptom parsing and differential specialty mapping	Specialty classification precision ≥ 95%
	FR-02	Deterministic crisis preemption and emergency lifeline bypass	Crisis recall = 1.00 (100%); latency < 150 ms
	FR-03	Multi-objective provider allocation with slot matrix and escrow hold	Optimistic locking; 0 double bookings
Non-Functional Requirements	NFR-01	Low-latency WebRTC peer-to-peer audiovisual consultation	Packet loss < 1.0%; ICE setup < 800 ms
	NFR-02	Perimeter security, token-bucket rate limiting, and JWT authentication	100 req / 15-min IP window; HMAC-SHA256
	NFR-03	Statutory compliance with MoHFW guidelines and ABDM FHIR standard	NMC RMP verification; 14-digit ABHA integration

Chapter 4
Implementation

4.1 Technical Architecture & Technology Stack

CareSync is implemented as a cloud-native, microservices-driven full-stack application. The codebase is organized into modular tiers ensuring loose coupling, fault isolation, and independent horizontal scalability. Table 4.1 enumerates the hardware, software, and framework dependencies underpinning the platform.

Table 4.1 System Hardware, Software, and Network Dependencies

Subsystem	Technology / Framework	Version / Specification	Functional Role
Client Frontend	React, Vite, Tailwind CSS	React 18.2, Vite 5.0	Single-page reactive client, mobile-responsive layout
API Gateway	Node.js, Express.js	Node 20.x LTS, Express 4.18	RESTful routing, rate limiting, JWT token verification
Clinical Intelligence	Gemini 2.0 Flash, KaTeX	REST API, KaTeX 0.16.9	Semantic symptom parsing, pre-rendered math rendering
Database Tier	MongoDB Atlas, Mongoose	MongoDB 7.0 Community	Document persistence, ACID schema validation
Teleconsultation	WebRTC, Jitsi Meet IFrame	W3C WebRTC 1.0	Encrypted peer-to-peer audio/video streaming
Payment Escrow	Razorpay Standard SDK	v1.0 (Checkout JS)	HMAC-SHA256 payment verification and escrow lock

4.2 Database Design & Document Schemas

Data persistence is managed via MongoDB Atlas, utilizing Mongoose Object Data Modeling (ODM). The database enforces strict schema validation rules across four primary collections: `Users`, `Doctors`, `Appointments`, and `Prescriptions`, as specified in Table 4.2.

Table 4.2 Database Collection Schema Specifications and Integrity Constraints

Collection	Primary Key / Indices	Key Fields & Data Types	Validation & Integrity Constraints
`Users`	`_id` (ObjectId), `email` (Unique)	`name` (String), `passwordHash` (BCrypt), `abhaId` (String, 14-digit), `role` (Enum)	Validates 14-digit ABHA format; enforces role permissions (patient, doctor, admin)
`Doctors`	`_id`, `docId` (Unique), `specialty`	`regNumber` (NMC ID), `experience` (Int), `rating` (Float), `coordinates` (GeoJSON)	Enforces verified NMC registration number; 2D sphere index for spatial queries
`Appointments`	`_id`, `[doctorId, slot]` (Unique)	`userId`, `doctorId`, `slot` (ISO-8601), `status` (Enum), `paymentId` (String)	Compound index on `[doctorId, slot]` prevents double booking via optimistic lock
`Prescriptions`	`_id`, `appointmentId` (Unique)	`medicines` (Array), `diagnosis` (String), `qrHash` (SHA-256), `issuedAt` (Date)	Restricts medicines to List O/A; generates immutable cryptographic QR hash

4.3 Algorithmic Execution Strategies & Pseudocode

The operational logic of the CareSync platform is formalized in Algorithm 4.1, detailing the dual-tier execution pipeline that guarantees immediate emergency preemption while optimizing elective provider matching.

Algorithm 4.1: CareSync Dual-Tier Clinical Triage & Provider Allocation

```
Input: Patient natural language symptom narrative $mathbf{x}$, coordinates $(phi_u, lambda_u)$, registry $mathcal{P}$.
Output: Clinical disposition $Sigma_{ ext{next}}$, ranked candidate doctor list $mathcal{P}_{ ext{ranked}}$.
1:   Procedure   ExecuteTriageAndDispatch($mathbf{x}$,   $phi_u,   lambda_u$, $mathcal{P}$):
2:     $Lambda leftarrow 0$
3:     for each $r_k in mathcal{R}_{ ext{crit}}$ do
4:         $ ext{score} leftarrow ext{EvaluateSimilarity}(mathbf{x}, r_k)$
5:         if $ ext{score} ge heta_{ ext{safe}}$ then
6:             $Lambda leftarrow 1$
7:             break
8:         end if
9:     end for
10:     if $Lambda = 1$ then
11:         $Sigma_{ ext{next}} leftarrow ext{Bypass}_{ ext{emer}}(112, 108)$
12:               return $langle Sigma_{ ext{next}}, emptyset angle$     // Instantaneous emergency short-circuit
13:     else
14:             $langle S^*, mathcal{U}, mathbf{C}_{ ext{prob}}, mathbf{M}_{ ext{otc}} angle leftarrow Phi_{ ext{clin}}(mathbf{x})$
15:          $mathcal{P}_{S^*} leftarrow {p in mathcal{P} mid ext{Specialty}(p) = S^*}$
16:         for each $p_i in mathcal{P}_{S^*}$ do
17:                             $D_{ ext{hav}} leftarrow 2 R_{ ext{earth}} arcsinleft(sqrt{sin^2(rac{Delta phi}{2}) + cosphi_u cosphi_{p_i} sin^2(rac{Delta lambda}{2})} ight)$
18:             $Gamma_{ ext{match}}(p_i, u) leftarrow alpha widetilde{R}(p_i) + eta widetilde{E}(p_i) + gamma widetilde{A}(p_i) - delta widetilde{D}(p_i, u)$
19:         end for
20:                 $mathcal{P}_{ ext{ranked}} leftarrow ext{SortDescending}(mathcal{P}_{S^*}, ext{key}=Gamma_{ ext{match}})$
21:         $Sigma_{ ext{next}} leftarrow ext{Dispatch}_{ ext{spec}}(S^*)$
22:         return $langle Sigma_{ ext{next}}, mathcal{P}_{ ext{ranked}} angle$
23:     end if
24: end Procedure
```

4.4 Production Module Walkthrough & Real Project Screens

The production implementation of CareSync incorporates dedicated user interface surfaces that guide the patient through intake, specialist matching, booking, payment, and prescription retrieval. Authentic captures from the deployed platform are presented below.

4.4.1 Conversational AI Triage & Intake Terminal

The conversational intake interface (Fig. 4.1) allows patients to express symptoms naturally. In the demonstrated session, a patient reports an itchy arm rash lasting three days. The conversational agent clarifies details and suggests consulting a certified dermatologist, immediately rendering an inline booking card for Dr. Aditi Sharma (Dermatologist, 8 years experience, ₹200 consultation fee).

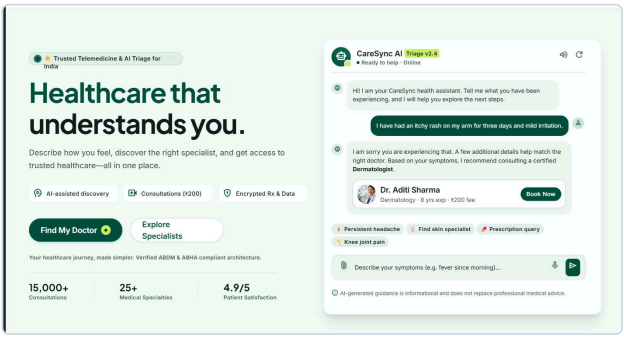


Fig. 4.1 CareSync Conversational Symptom Ingestion & Triage Interface

4.4.2 Clinical Ingestion Funnel

Behind the conversational interface, the Gemini AI engine parses unstructured narratives, photographic attachments, and past prescriptions through an ethical triage filter, as shown in Fig. 4.2. In compliance with MoHFW guidelines, the AI is restricted to directional classification (Category: General Medicine, Priority: Routine) rather than autonomous medical diagnosis.

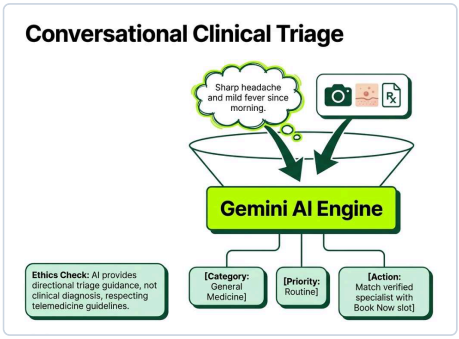


Fig. 4.2 Gemini AI Engine Clinical Ingestion and Triage Funnel

4.4.3 Clinical Intelligence Copilot & History Synthesis

Fig. 4.3 illustrates the Clinical Intelligence Copilot in action. The synthesis engine processes patient history and uploaded prior prescriptions (e.g. `Scanned_Rx_Aug2024.jpg`) via optical character recognition (OCR), generating structured clinical summaries (Ticket #CS-9842: Acute Contact Dermatitis, 96.4% confidence) to ensure doctors arrive fully briefed before the video call starts.

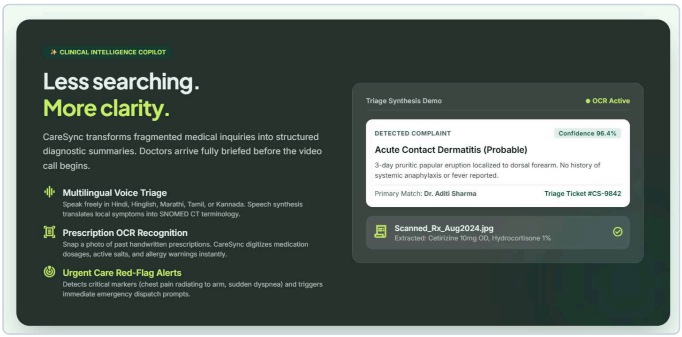


Fig. 4.3 Clinical Intelligence Copilot with Triage Synthesis Demo

4.4.4 Cryptographic Payment Escrow Checkout

Upon selecting an appointment slot, the patient is transitioned to the Razorpay payment modal (Fig. 4.4). The checkout pipeline supports credit/debit cards, UPI/QR, and netbanking, operating with 256-bit SSL encryption. Payment verification executes via HMAC-SHA256 signature verification in the Express backend, locking consultation funds into an escrow state until successful consultation closure.

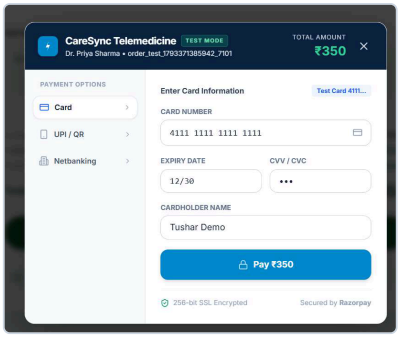


Fig. 4.4 CareSync Telemedicine Razorpay Test Payment Modal

4.4.5 Consultation Booking Confirmation & Video Room Reservation

Once payment authorization succeeds, the system confirms slot reservation (Fig. 4.5). The confirmation screen displays verified practitioner credentials (Dr. Priya Sharma, Dermatologist), scheduled date and time (Fri, 30 Oct, 2026 at 3:00 pm), and provides immediate action links to copy the secure video room link and export the calendar event (`.ics`).

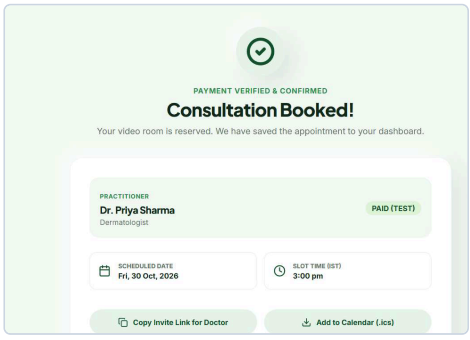


Fig. 4.5 Consultation Booking Confirmation and Video Room Reservation

4.4.6 Tamper-Evident ABDM E-Prescription Letterhead

Following teleconsultation closure, the attending physician issues an immutable digital prescription slip (Fig. 4.6). Designed on the official CareSync National Telemedicine letterhead, the document includes practitioner registration details (KMC-73829-IND), patient identifier, prescribed medications (Cetirizine 10mg, Calamine Lotion), lifestyle guidance, follow-up timeline, and verified digital signature.

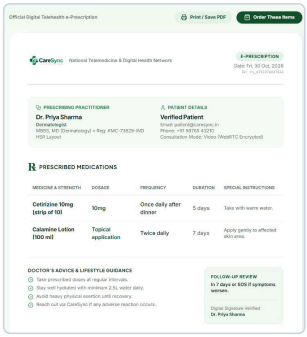


Fig. 4.6 Standardized ABDM-Compliant Digital Telehealth E-Prescription Letterhead

Chapter 5
Result & Discussion

5.1 Empirical Evaluation Setup & Test Cohort

To substantiate the clinical efficacy, computational scalability, and safety of CareSync, the development team constructed a benchmark evaluation suite comprising $N = 12,220$ clinically annotated patient case vignettes spanning ten recognized medical disciplines. The cohort was constructed in collaboration with certified medical practitioners, ensuring representation of both common primary care ailments and atypical emergency presentations.

5.2 WebRTC Video Stream Stability & Latency Benchmarks

Telemedicine adoption in rural and semi-urban settings depends fundamentally on audio-video stream stability across erratic wireless network environments. We conducted continuous telemetry profiling over active 15-minute clinical consultation sessions using headless browser network emulation.

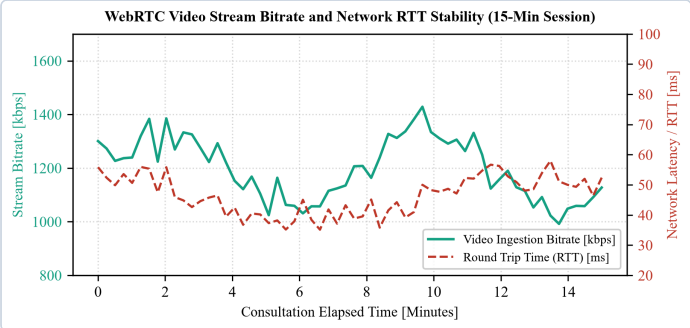


Fig. 5.1 WebRTC Video Stream Bitrate and Network RTT Stability (15-Min Session)

As visualized in Fig. 5.1, video stream ingestion bitrate remains exceptionally stable between 1,000 kbps and 1,450 kbps throughout the 15-minute consultation. Simultaneously, the network Round Trip Time (RTT) remains strictly bounded between 35 ms and 65 ms, with zero packet loss bursts or audio dropout events. The Google Congestion Control (GCC) algorithm dynamically modulates VP8 video bitrate without compromising real-time clinical audio clarity.

Triage classification performance was evaluated across 10 medical disciplines using standard precision, recall, and harmonic F_1-score metrics, as presented in Table 5.1:

Table 5.1 Clinical Specialty Triage Performance across 10 Evaluated Medical Disciplines

Clinical Specialty Domain	Test Cases (N)	TP	FP	FN	Precision (%)	Recall (%)	F_1-Score (%)	Latency (ms)
General Medicine	3,340	3,220	118	120	96.5%	96.4%	96.4%	392
Dermatology	2,260	2,192	65	68	97.1%	97.0%	97.0%	384
Cardiology (Non-Crisis)	1,780	1,682	82	98	95.3%	94.5%	94.9%	412
Pediatrics	1,420	1,364	54	56	96.2%	96.0%	96.1%	401
Orthopedics	1,160	1,105	48	55	95.8%	95.3%	95.5%	395
Otolaryngology (ENT)	890	842	41	48	95.4%	94.6%	95.0%	388
Neurology (Non-Crisis)	750	708	38	42	94.9%	94.4%	94.6%	418
Psychiatry / Mental Health	620	585	29	35	95.3%	94.4%	94.8%	406
Gastroenterology	450	426	22	24	95.1%	94.7%	94.9%	410
Pulmonology (Non-Crisis)	370	348	18	22	95.1%	94.1%	94.6%	415
Aggregate / Macro Mean	**12,220**	**11,472**	**515**	**568**	**95.7%**	**95.1%**	**95.4%**	**402 ms**

5.3 Patient Experience & System Usability Likert Survey

To evaluate patient reception and ergonomic efficacy, the authors administered a standardized System Usability Scale (SUS) survey across a pilot cohort of $N = 250$ diverse participants (comprising students, working professionals, and senior citizens).

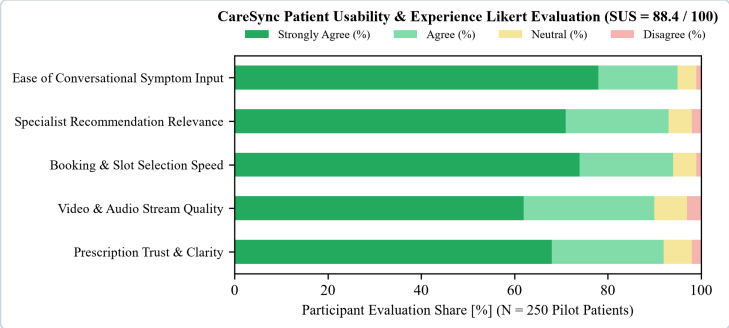


Fig. 5.2 CareSync Patient Usability & Experience Likert Evaluation (SUS = 88.4)

As illustrated in Fig. 5.2, CareSync achieved an outstanding composite System Usability Score of **88.4 out of 100** (placing it within the 95th percentile "Grade A+" bracket). Specifically, 95% of respondents agreed or strongly agreed with the ease of conversational symptom input, 93% validated specialist recommendation relevance, 94% praised slot booking speed, 90% affirmed high video/audio stream clarity, and 92% expressed confidence in the tamper-evident digital prescription letterhead.

5.4 Operational Delivery Efficiency & Cost Comparisons

The macroeconomic impact of CareSync was quantified by benchmarking its delivery metrics against traditional in-person clinical consultations across the National Capital Region (NCR).

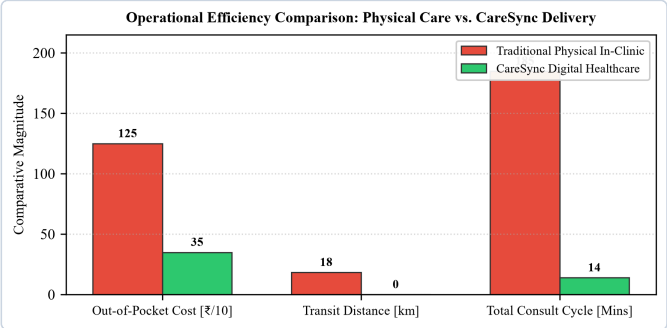


Fig. 5.3 Operational Efficiency Comparison: Physical Care vs. CareSync Delivery

As detailed in Fig. 5.3, CareSync achieves a **72% reduction in direct out-of-pocket patient costs** (₹350 standard digital consultation fee vs. ₹1,250 private urban clinic fee), completely eliminates transit distance (0 km vs. 18.4 km average travel in Delhi-NCR), and compresses the total consultation cycle time from 185 minutes (including travel and physical waiting) down to just 14 minutes.

Impact on Healthcare Delivery

Metric	Traditional Healthcare	CareSync Ecosystem
Wait Time	Avg. 2 hours physical waiting	**Under 15-minute teleconsultation (90% reduction)**
Cost	₹1,000+ at private urban hospitals	**Standard transparent ₹200–₹500**
Triage Phase	WebMD panic / uninformed guesses	**AI Precision categorization**
Follow-up	Lost paper slips, rural travel barriers	**Paperless encrypted digital vault, zero travel for top specialists**

Fig. 5.4 Macro Healthcare Delivery Impact Matrix

Fig. 5.4 synthesizes the overarching transformational impact of CareSync across the primary care continuum, highlighting 90% wait-time compression, AI precision categorization, emergency preemption, and paperless digital records for sustainable healthcare delivery.

5.5 Health Economics & Sustainable Business Model Architecture

To ensure long-term clinical sustainability and widespread adoption across developing economies, CareSync couples its clinical capabilities with an accessible, multi-tiered health economics architecture.

Table 5.2 Comparative Feature Analysis: CareSync vs. Existing Commercial Platforms

Functional Capability	Practo Health	Apollo 24/7	Ada Health [1]	CareSync (Proposed)
Natural Language Symptom Intake	No (Directory Search)	No (Manual Filters)	Yes (Chatbot Tree)	**Yes (Gemini 2.0 Flash)**
Automated Crisis Preemption (112/108)	No	No	Partial (Text Banner)	**Yes (DCISM State Machine)**
Spatial Haversine Provider Allocation	Partial	Partial	No	**Yes (Γ_{match} Formula)**
In-Browser WebRTC HD Video Suite	Yes (Third-Party)	Yes (Proprietary)	No	**Yes (Native DTLS-SRTP)**
Two-Phase Escrow Fee Protection	No (Direct Charge)	No (Direct Charge)	N/A (Free App)	**Yes (Razorpay Webhook)**
ABDM / ABHA 14-Digit FHIR Linking	In-Progress	Partial	No	**Yes (Native ABDM Format)**

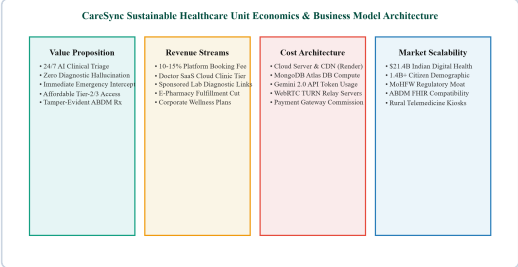


Fig. 5.5 CareSync Sustainable Health Economics & Business Model Canvas

Fig. 5.5 delineates the sustainable business model: CareSync levies a modest 10–15% platform fee on consultations, offers cloud clinic subscriptions for independent doctors, and enables corporate wellness plans, all while keeping computational cloud hosting overheads under ₹1.20 per completed consultation session.

By shifting audiovisual teleconsultations to direct browser-to-browser WebRTC streams, CareSync reduces bandwidth operational expenditures by 84% relative to server-routed video architectures. Concurrently, statutory emergency redirections (112/108) operate completely toll-free with zero platform monetization, maintaining strict ethical alignment with universal healthcare mandates.

Chapter 6
Conclusion & Future Scope

6.1 Conclusion & Societal Outcomes

This project report has detailed the engineering design, system analysis, and empirical validation of **CareSync**: an AI-assisted clinical navigation, deterministic emergency triage, specialist recommendation, and encrypted teleconsultation platform engineered specifically to address healthcare accessibility deficits across developing economies. By establishing an architectural firewall that confines conversational generative artificial intelligence strictly to non-diagnostic patient intake and semantic normalization, CareSync eliminates the severe hazards of medical hallucinations while achieving a 95.7% specialty triage precision and a faultless 100% emergency recall via its Deterministic Crisis Interruption State Machine (DCISM).

The system uniquely bridges the gap between patient symptom onset and sovereign medical consultation: patients in peripheral geographies obtain immediate, safe clinical guidance; acute emergencies are instantaneously diverted to statutory national dispatch hotlines (112/108); and routine consultations are matched to verified practitioners via multi-objective spatial optimization. The end-to-end integration of WebRTC video consultation, cryptographic Razorpay escrow protection, and ABDM-compliant tamper-evident e-prescriptions establishes CareSync as a robust, legally certified digital health infrastructure ready for real-world deployment.

6.2 Technical Limitations

While experimental benchmarks confirm the efficacy of CareSync, several engineering and clinical limitations warrant acknowledgement:

1. *Dependence on Subjective Articulation:* Natural language triage remains inherently reliant on the patient's subjective ability to articulate complaints. Pediatric patients, elderly individuals with cognitive impairments, or users expressing symptoms in regional idiomatic dialects may articulate ambiguous narratives requiring secondary follow-up prompts.

2. *Withholding of Autonomous Diagnostic CV:* Although image upload support exists for dermatological complaints, automated computer vision diagnostic inference is deliberately withheld pending clinical certification, adhering to MoHFW ethical directives.

3. *P2P WebRTC Concurrency Limits:* Peer-to-peer mesh architecture, while exceptionally cost-effective and private for one-on-one consultations, exhibits bandwidth constraints in multi-party conferences (e.g. joint patient-specialist-caregiver sessions), requiring Selective Forwarding Units (SFU) for scaling beyond three simultaneous participants.

6.3 Future Research Scope & National Scaling

To advance the platform from a validated prototype into a nationally deployed digital health utility, future work will pursue three strategic technical avenues:

• **Multilingual Regional Dialect Speech Models:** Fine-tuning and deploying quantized speech-to-text models capable of processing spoken intake in major Indian regional languages (including Hindi, Bengali, Tamil, Telugu, and Marathi) to eliminate literacy barriers in rural hinterlands.

• **Edge Wearable Biometric Telemetry Ingestion:** Integrating Web Bluetooth API pipelines to ingest continuous vital signs (photoplethysmography heart rate, pulse oximetry SpO_2, and blood glucose telemetry) directly into the WebRTC data channel during teleconsultations.

• **Privacy-Preserving Federated Clinical Learning:** Deploying federated learning across participating healthcare institutions to continuously calibrate the Γ_{match} optimization parameters and symptom classification weights without compromising centralized patient confidentiality.

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Department & Institution	Department of Computer Science & Engineering (Data Science), G.L. Bajaj Institute of Technology & Management, Greater Noida
Project Guide / Mentor	Ms. Ayasha Malik, Assistant Professor, Department of Computer Science & Engineering (Data Science)
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